**Code No: 114AG** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech II Year II Semester Examinations, September/October - 2023**
**FORMAL LANGUAGES AND AUTOMATA THEORY**
**(Computer Science and Engineering)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

JNTUH USED PAPERS-2023

## PART - A

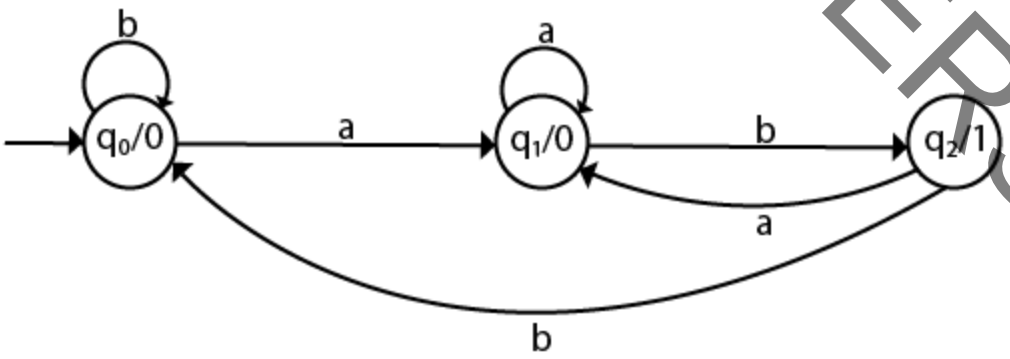
**(25 Marks)**

1.a) What is the acceptance of a string in automata? [2]
b) Explain NFA with ε transitions with example. [3]
c) Identify the properties of Derivation Tree. [2]
d) Write the regular expression for the language accepting all combinations of a's except the null string, over the set $\sum = \{a\}$. [3]
e) Identify the applications of Context Free grammar. [2]
f) Construct a CFG for the regular expression (0+1)*. [3]
g) Identify the components of Turing Machine. [2]
h) Explain about counter machine. [3]
i) Define P problems. [2]
j) Write about Universal TM. [3]

## PART - B

**(50 Marks)**

2.a) Compare DFA and NFA and quote examples for both.
b) Convert the given Moore machine into its equivalent Mealy machine. [5+5]




**OR**

3.a) Discover the ground rules of finite automata.
b) Identify steps for minimization of DFA with an example. [5+5]

4.a) Construct the FA for regular expression 0*1 + 10.
b) Explain Right and left Linear Regular grammars with example. [5+5]

**OR**

5.a) Illustrate the rules and properties of regular expression.
b) Write a detail note on pumping lemma of regular expressions. [5+5]

6.a) Design a Push Down Automata for accepting a language {anb2n | n>=1}.
b) Explain Greibach Normal Form with example. [5+5]

**OR**

7.a) Check whether the given grammar G is ambiguous or not.

$A \rightarrow AA$
$A \rightarrow (A)$
$A \rightarrow a$

b) Identify the conventional notations of Push Down Automata. [5+5]

8.a) Construct TM for the addition function for the unary number system.
b) Examine linear bounded automata with example. [5+5]

**OR**

9.a) Illustrate the Properties of Context Sensitive Language.
b) Discuss Church's Hypothesis in detail. [5+5]

10.a) Prove that halting problem is undecidable.
b) Explain post correspondence problem with an example. [5+5]

**OR**

11.a) Difference between NP hard Problem and NP complete Problem.
b) State when a problem is said to be decidable and give an example for decidable problem. [5+5]

**---ooOoo---**

JNTUH USED PAPERS-2023